Parsanna Koirala

+1 (240)-817-9421 parsanna.koirala@gmail.com
linkedin.com/in/parsanna-koirala-931b5b250/ github.com/ParsannaK

Summary

Software engineer with hands-on experience in software development, AI, and computer vision. Skilled in Python, C++, Swift, JavaScript, and more, with a track record of building scalable tools and machine learning models. Seeking to contribute to impactful engineering projects through innovative solutions and strong problem-solving skills.

Education

University of Michigan - Ann Arbor **Aug 2022 – Dec 2025**
Bachelor of Science, Computer Science

Relevant Coursework

- EECS 442: Computer Vision
- EECS 445: Machine Learning
- EECS 388: Computer Security
- EECS 370: Computer Organization
- LING 441: Computational Linguistics
- EECS 481: Software Engineering
- EECS 281: Data Structures and Algorithms

Experience

Software Engineer & Product Management Intern **Jan 2026 – Present**
ECS Fin *New York City, NY*

- Developed scalable financial messaging components using Java EE and enterprise design patterns, strengthening skills in resilient application architecture and distributed system thinking.
- Partnered with technical teams and stakeholders to translate complex business requirements into actionable technical specifications, user stories, and implementation plans.
- Built and optimized SQL queries, stored procedures, and triggers to support high-volume transaction workflows across Oracle and SQL Server environments.
- Collaborated in Agile development cycles, performing code reviews and testing to improve software quality, maintainability, and production reliability.
- Supported product planning by incorporating stakeholder and customer feedback into feature prioritization and roadmap discussions.
- Trained on international financial messaging standards, including ISO 20022, SWIFT, and SEPA, to ensure software compliance with global banking regulations.

Lead Developer **Jan 2023 – Dec 2025**
University of Michigan Autonomous Robotic Vehicle Team *Ann Arbor, MI*

- Led development of deep learning models for lane detection and driveable area prediction, applying AI/ML techniques to solve real-time perception problems.
- Designed and optimized DCNN and DRNN architectures for live video streams, achieving 87% accuracy in driveable area prediction for autonomous vehicle competition use.
- Improved model performance through experimentation, evaluation, and iterative system refinement under real-world constraints.

Building Supervisor **Apr 2024– Dec 2025**
University of Michigan - Ann Arbor *Ann Arbor, MI*

- Led a team of 20 staff members in a fast-paced customer-facing environment, ensuring smooth operations, issue resolution, and high-quality service.
- Communicated clearly with diverse stakeholders to resolve scheduling, facility, and member concerns efficiently.
- Strengthened leadership, customer advocacy, and operational problem-solving skills through day-to-day team and facility management.

Core Skills

Languages: Python, C++, JavaScript, Swift, Java, Node.js, C#
Cloud/Architecture: Scalable System Design, Resilient Application Design, Technical Solution Design, Cloud Concepts, API Integration
AI/ML: Machine Learning, Deep Learning, Computer Vision, CoreML, OpenCV
Frameworks & Tools: React.js, Electron, Git, Maven, SQL, Oracle, IntelliJ, Eclipse, SwiftUI
Other: SQL, R, SAS, Hadoop, Spark, Tableau, SPSS, VBA, HTML, Databases
AI & Engineering Practices: Stakeholder Communication, Requirements Gathering, Agile Development, Customer-Focused Problem Solving, Data Structures & Algorithms

Projects

FloraGuide: AI Garden Assistant |*React Native, Expo, TypeScript, Supabase* **Jan 2026 – Present**
Solo Developer

- Architected and deployed a cross-platform mobile application for iOS and Android using React Native and Expo, maintaining a single TypeScript codebase to ensure consistent UI/UX across platforms.
- Integrated Google Gemini AI via a custom Node.js middleware to provide real-time plant identification and an interactive gardening assistant that contextualizes advice based on user-specific growth journals.
- Developed a resilient data synchronization layer using Supabase and TypeScript, enabling secure user authentication and real-time cloud storage for plant records, high-resolution images, and care schedules.
- Engineered custom native modules for local notifications and camera integration, automating plant care workflows including species-specific watering, fertilizing, and pruning reminders.
- Optimized application performance by managing the JavaScript heap and implementing efficient asset scaling, resulting in a responsive experience even during high-compute AI processing tasks.
- Managed the full DevOps lifecycle using EAS and Fastlane to automate local and production builds, navigating the App Store submission process and AI disclosure regulations.

My2Do - iOS Productivity App |*Swift, SwiftUI, CoreML, SwiftData* **Oct 2025 – Dec 2025**
apps.apple.com/us/app/my2do-tasks-habits/id6756674181

- Engineered and published a native iOS productivity app to the App Store, serving 1,000+ active downloads through a polished and scalable mobile experience.
- Integrated external APIs and system frameworks including weather services, scheduling tools, and local notifications to support real-time, user-centered functionality.
- Designed reusable app features and data flows for task scheduling, habit tracking, and workflow customization.

Computer Setup Assistant App |*React.js, Node.js Figma, Electron* **Jan 2025 – Apr 2025**
Software Design and Development Group

- Collaborated in a 4-person team to design and build a responsive web app that guided new users through software installation and system configuration.
- Implemented step-by-step interactive workflows, using React for the frontend and Node.js for handling user state.

Languages

English (fluent), **Nepali** (fluent), **Spanish** (intermediate)